

# RAYA DAS

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LinkedIn — GitHub — Portfolio — Leetcode

## PROFESSIONAL SUMMARY

Computer Science Engineering graduate skilled in HTML, CSS, JavaScript, React.js, SQL, REST APIs, and Machine Learning fundamentals. Strong foundation in DSA, OOP, DBMS, OS, and Computer Networks, with hands-on experience building real-world projects. Eager to contribute technical skills, problem-solving abilities, and a passion for continuous learning in a dynamic technology-driven environment.

## TECHNICAL SKILLS

**Programming Languages:** Python, Java, JavaScript, SQL

**Web Development:** HTML5, CSS3, React.js, REST APIs

**Machine Learning:** PyTorch, Supervised Learning, Data Preprocessing, Model Evaluation

**Databases:** PostgreSQL, SQL

**Tools & Platforms:** Git, GitHub, Postman, Streamlit, Linux, AWS (EC2, S3)

**AI-Assisted Development:** Prompt Engineering, AI Workflows, LLM-Assisted Development, Code Generation & Debugging

**Core Computer Science:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks

## RESEARCH EXPERIENCE

**Emotify: Design of a Multimodal Emotion Detection System**

Sept 2025

Presented at the 3rd International Conference on Computational Applied Sciences and its Applications (ICCASA-2025)

- Co-authored and presented a research paper proposing a multimodal deep learning framework combining CNN and BiLSTM models for emotion recognition.
- Developed a multimodal fusion approach integrating facial and speech features to improve emotion classification performance.
- Explored practical applications of multimodal emotion detection in human-computer interaction systems.

## PROJECTS

**Multimodal Emotion Detection System**

- Built a multimodal emotion recognition system by combining facial (ResNet-50) and speech (BiLSTM) features through feature-level fusion.
- Achieved **85% accuracy**, outperforming unimodal models by **25%** and improving emotion classification across 7 emotion classes.
- Developed a real-time Streamlit application for automated emotion prediction from image and audio inputs.

**Ad Campaign Analytics Dashboard**

GitHub

- Developed a full-stack analytics dashboard to track CTR, ROAS, impressions, clicks, and campaign spending metrics.
- Designed REST APIs and optimized PostgreSQL queries, reducing data retrieval time by approximately **30%**.
- Implemented campaign reporting and visualization features, enabling centralized monitoring across multiple campaigns.

**AI-Powered Newsletter Generator**

GitHub

- Built an automated newsletter generation system integrating multiple third-party APIs for content aggregation and processing.
- Reduced manual newsletter creation effort by over **70%** through workflow automation and asynchronous API orchestration.
- Improved reliability with response validation, error handling, and structured processing pipelines.

## EDUCATION

**Bachelor of Technology (B.Tech) in Computer Science and Engineering**

July 2026

University of Engineering and Management, Jaipur

CGPA: 7.68/10.0